

Document number: P3428R3

Date: 2026-05-11

Project: Programming Language C++, LWG

Authors: Maged M. Michael, Michael Wong, Paul McKenney

Email: maged.michael@acm.org, fraggamuffin@gmail.com, paulmck@kernel.org

Hazard Pointer Batches

Table of Contents

Introduction	2
Background: P2530R3 C++26 Hazard Pointers	2
Motivation	3
Implementation and Use Experience	3
Batches of Hazard Pointers	3
Proposed Wording	3
Usage Example	4
History	5
References	6

Introduction

This paper proposes extending the C++26 hazard pointer interface to support creation and destruction of batches of nonempty hazard pointers.

This revision P3428R3 revises R2 by following LEWG Croydon 2026 feedback. It is ready for review by LWG in Brno 2026.

Background: P2530R3 C++26 Hazard Pointers

Hazard pointer interface from P2530R3 merged into the working draft (N5008):

```
template <class T, class D = default_delete<T>>
class hazard_pointer_obj_base {
public:
    void retire(D d = D()) noexcept;
protected:
    hazard_pointer_obj_base() = default;
    hazard_pointer_obj_base(const hazard_pointer_obj_base&) = default;
    hazard_pointer_obj_base(hazard_pointer_obj_base&&) = default;
    hazard_pointer_obj_base& operator=(const hazard_pointer_obj_base&) = default;
    hazard_pointer_obj_base& operator=(hazard_pointer_obj_base&&) = default;
    ~hazard_pointer_obj_base() = default;
private:
    D deleter ; // exposition only
};

class hazard_pointer {
public:
    hazard_pointer() noexcept; // Constructs an empty hazard_pointer
    hazard_pointer(hazard_pointer&&) noexcept;
    hazard_pointer& operator=(hazard_pointer&&) noexcept;
    ~hazard_pointer();
    [[nodiscard]] bool empty() const noexcept;
    template <class T> T* protect(const atomic<T*>& src) noexcept;
    template <class T> bool try_protect(T*& ptr, const atomic<T*>& src) noexcept;
    template <class T> void reset_protection(const T* ptr) noexcept;
    void reset_protection(nullptr_t = nullptr) noexcept;
    void swap(hazard_pointer&) noexcept;
};

hazard_pointer make_hazard_pointer(); // Constructs a nonempty hazard_pointer
void swap(hazard_pointer&, hazard_pointer&) noexcept;
```

Brief notes (See P2530R3 for details):

- A `hazard_pointer` object is either *empty* or *nonempty*. It is nonempty if and only if it owns a hazard pointer. Only nonempty `hazard_pointer` objects can be used to protect protectable objects.
- The default constructor of `hazard_pointer` constructs an empty object, whereas the free function `make_hazard_pointer` constructs a nonempty object.

Motivation

The construction and destruction of multiple nonempty `hazard_pointer` objects in one batch has lower latency than their construction and destruction separately, e.g., 2 ns vs 6 ns for the construction/destruction of 3 nonempty hazard pointers.

Implementation and Use Experience

Batches of hazard pointers have been part of the Folly open-source library (under the name `hazptr_array` as a distinct class) and in heavy use in production since 2017.

Batches of Hazard Pointers

The construction and destruction of a nonempty `hazard_pointer` object typically involves access to thread-local storage and has low but non-negligible latency (low single digit nanoseconds). The P2530R3 C++26 hazard pointer interface supports only the construction and destruction of nonempty hazard pointers individually.

This paper proposes adding support for the construction and destruction of multiple nonempty hazard pointers in one batch to reduce the latency of such operations. This revision also adds a batch move function to make it easy for users to keep batches of hazard pointer objects empty or nonempty together.

Proposed Wording

Edit **17.3.2 [version.syn] p2** as follows:

`__cpp_lib_hazard_pointer` ~~202306L~~ 202XXXXL

[Editor's note: The value will be set to the date of the meeting where this paper is approved for the working draft.]

Add the following to **32.11.3.4.4 [saferecl.hp.holder.nonmem]**:

```
void make_hazard_pointer_batch(std::span<hazard_pointer> span);
```

Effects: If an exception is thrown, there are no effects. Otherwise, for each element `e` of `span` such that `e.empty()` is true, constructs a hazard pointer and makes `e` the owner of that hazard pointer.

Postconditions: `e.empty()` is false for each element `e` of `span`.

Throws: `bad_alloc` if memory for any of the newly-constructed hazard pointers could not be allocated.

```
void clear_hazard_pointer_batch(std::span<hazard_pointer> span) noexcept;
```

Effects: For each element *e* of *span*, if *e.empty()* is false, destroys the hazard pointer owned by *e* and makes *e* empty.

Postconditions: *e.empty()* is true for each element *e* of *span*.

Usage Example

The following table shows two functionally-equivalent code snippets using the P2530R3 C++26 hazard pointer interface and using hazard pointer batches.

P2530R3 C++26	Hazard Pointer Batches
<pre>{ hazard_pointer hp[3]; /* Three hazard pointers are made nonempty separately. */ hp[0] = make_hazard_pointer(); hp[1] = make_hazard_pointer(); hp[2] = make_hazard_pointer(); assert(!hp[0].empty()); assert(!hp[1].empty()); assert(!hp[2].empty()); // Use the hazard pointers as usual // src is atomic<T*> T* ptr = hp[0].protect(src); /* etc */ } /* Three nonempty hazard pointers are destroyed separately. */</pre>	<pre>{ hazard_pointer hp[3]; /* Three hazard pointers are made nonempty together. */ make_hazard_pointer_batch(hp); assert(!hp[0].empty()); assert(!hp[1].empty()); assert(!hp[2].empty()); // Use the hazard pointers as usual // src is atomic<T*> T* ptr = hp[0].protect(src); /* etc */ clear_hazard_pointer_batch(hp); /* Three nonempty hazard pointers are emptied together. */ } /* The three emptied hazard pointers are destroyed separately. */</pre>

History

The Varna 2023 plenary voted in favor of including hazard pointers in the C++26 standard library ([2023-06 LWG Motion 7] P2530R3 Hazard Pointers for C++26).

P3135R1 was presented to SG1 in Tokyo 2024, reviewing potential extensions of the P2530R3 C++26 interface, and proposing two of those for inclusion in the standard library. The proposal for extending the P2530R3 C++26 interface to support batch creation and destruction of nonempty hazard pointers was voted on by the SG1 in Tokyo 2024 with unanimous consent:

we want to continue work on hazard pointer batches for C++26, except as free-functions that take and return an existing type of collection (e.g. `std::array`, or a range, or...)

R0

P3428R0 was a follow up on P3135R1, focusing on extending the C++26 hazard pointer interface to support batch creation and destruction of nonempty hazard pointers, revised to take into account SG1 feedback:

- Use free-functions that take and return an existing type of collection (e.g. `std::array`, or a range, or...) instead of a new class.

R1

R1 added draft wording to R0. D3428R1 was reviewed by SG1 in Wroclaw 2024 and SG1 voted with unanimous consent to forward (D)P3427R1 to LEWG for C++26 with feedback:

Forward (D)P3428R1 to LEWG for C++26 with notes:
* The preconditions don't have to be this strict
* The names `reset` and `move` can be changed by LEWG

There was also feedback to change the parameter names `span1` and `span2` to `span_from` and `span_to`, and to replace `N` with `span.size()`.

R2

R2 revises R1 to follow the feedback from SG1 and for review by LEWG:

- Relax the preconditions and update the wording accordingly.
- Change `reset_hazard_pointer_batch` to `clear_hazard_pointer_batch`.
- Change `span1` and `span2` to `span_from` and `span_to`.
- Use `span.size()` instead of `N` where applicable.

R3

R3 revises R2, which was reviewed by LEWG in Croydon 2026.

POLL: Forward P3428R2 (add strong exception guarantee and feature test macro and remove `move_hazard_pointer_batch`) to LWG for C++29

R3 applies the feedback from LEWG and is ready for review by LWG:

- Bump feature test macro
- Remove `move_hazard_pointer_batch`.
- Strengthen exception guarantee and polish wording in general.

References

- [P2530R3](#): Hazard Pointers for C++26 (2023-03-02).
- [P3135R1](#): Hazard Pointer Extensions (2024-04-12).
- [Folly](#): Facebook Open-source Library.